Line

equation → $\boxed{ax + by + c = 0}$

⇓

$by = -ax - c$

$y = -\frac{a}{b}x - \frac{c}{b}$

⇓

Similar to

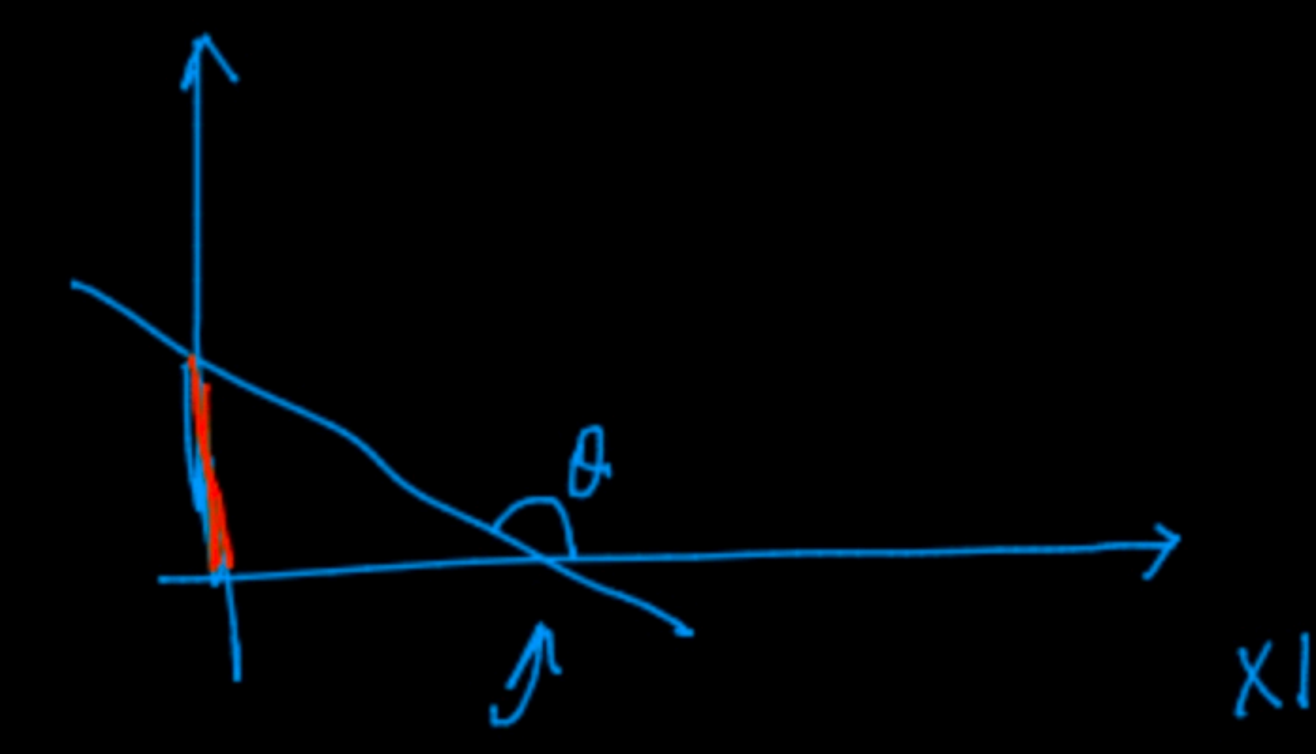
$Y = mx + c$

slope, y-intercept

$y = mx + c$

$\boxed{m = -\frac{a}{b}}$

eg.

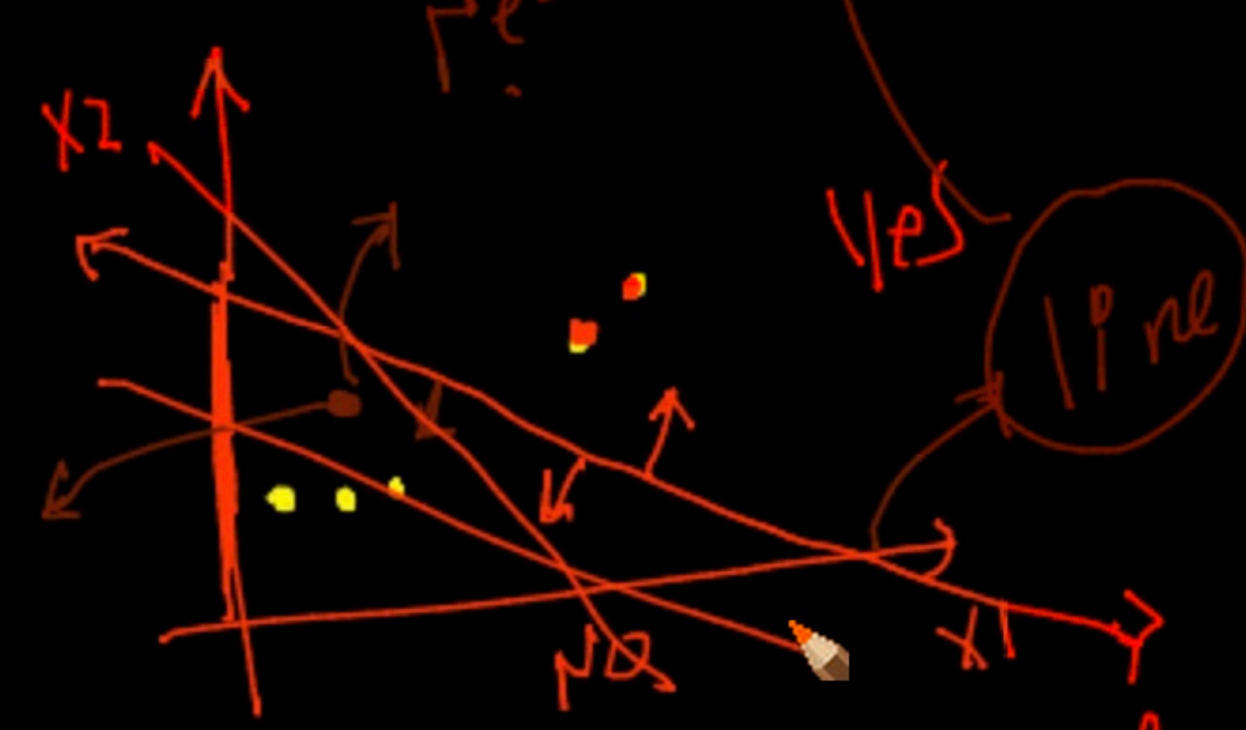


Decision Surface

eg. Dx

X1	X2	Stud
3	2	NO
3.2	2.5	NO
3.4	2.8	NO
5	7.5	Yes
6	8.5	Yes

plot ←



New
[3.8, 3.4]

No

$l \rightarrow m$, c

$w_1 * x_1 + w_2 \cdot x_2 + w_0 = 0$ ← equation of line

→ $\boxed{2D}$ → $\boxed{3D}$ →

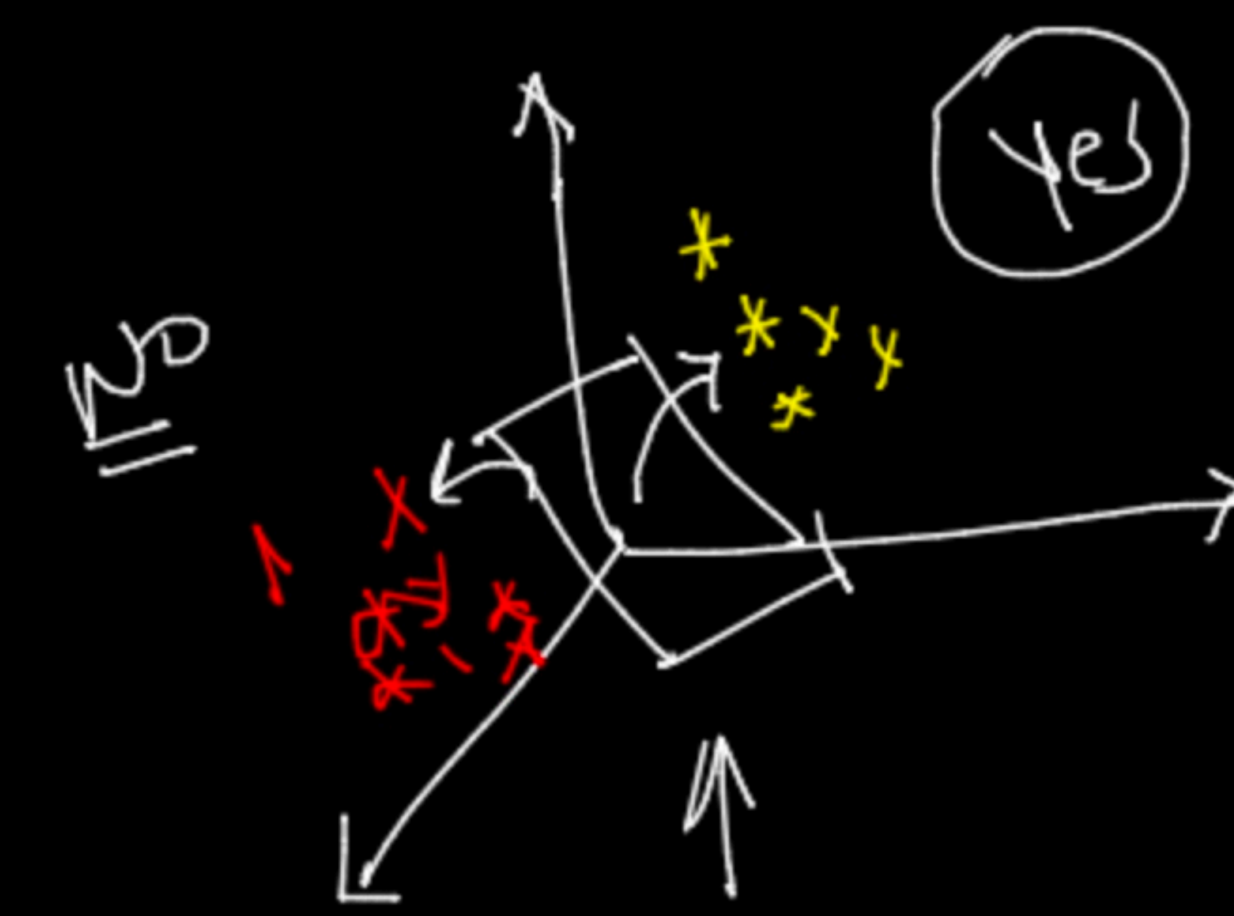
$\boxed{\text{line}}$ (plane)

equations

$x_1 \quad x_2 \quad x_3$

⇓

$\boxed{w_1 \cdot x_1 + w_2 x_2 + w_3 x_3 + w_0 = 0}$



plane

$\boxed{2D}$ → line

$\boxed{3D}$

x_1	x_2	x_3	~~x4~~ y	~~x5~~
—	—	—	y	
—	—	—	:	
—	—	—		

→ plot

$\boxed{nD}$ → Hyperplane

↓

$w_1 * x_1 + w_2 * x_2 + w_3 * x_3 + \ldots w_n \cdot x_n + w_0 = 0$

$$W_1 \cdot x_1 + W_2 \cdot x_2 + w_3 \cdot x_3 + \cdots w_n \cdot x_n + w_0$$

$$\boxed{\sum_{i=1}^{n} w^T \cdot x_i + w_0 = 0}$$ ← equation of Hyperplane

$$\begin{bmatrix} 0.1, & 0.2, & 0.3 \end{bmatrix} \to (1 \times 3)$$

$$\begin{array}{ccc} \uparrow & \uparrow & \uparrow \\ w_1 & w_2 & w_3 \end{array}$$

$$\begin{array}{ccc} X_1 & x_2 & X_3 \end{array}$$

$$\begin{bmatrix} 7 & 5 & 2 \end{bmatrix}$$

(1×3)

$$\begin{bmatrix} 0.1 \\ 0.2 \\ 0.3 \end{bmatrix} [7, 5, 2]$$

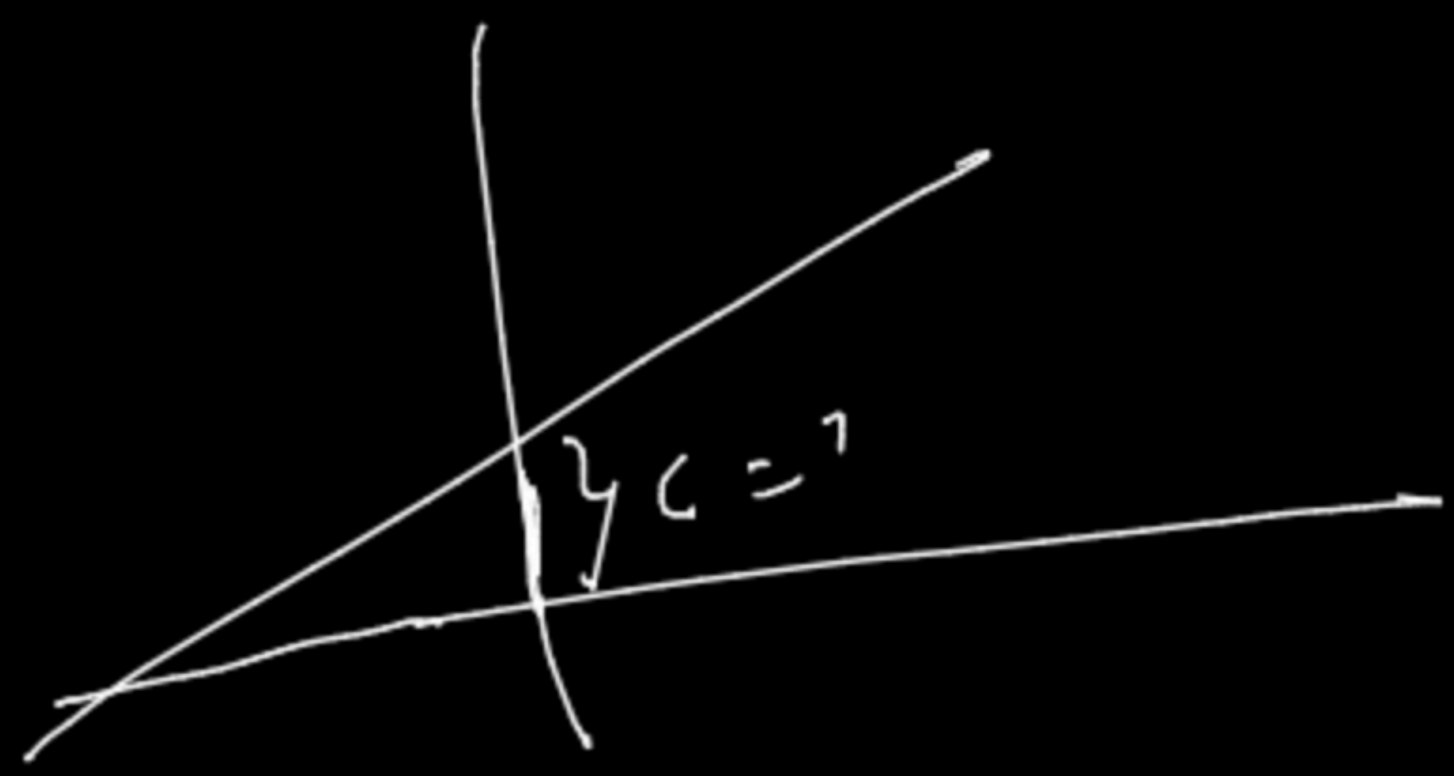
$$\Downarrow$$

$$0.1 \times 7 + 0.2 \times 5 + 0.3 \times 2$$

$$\Downarrow$$

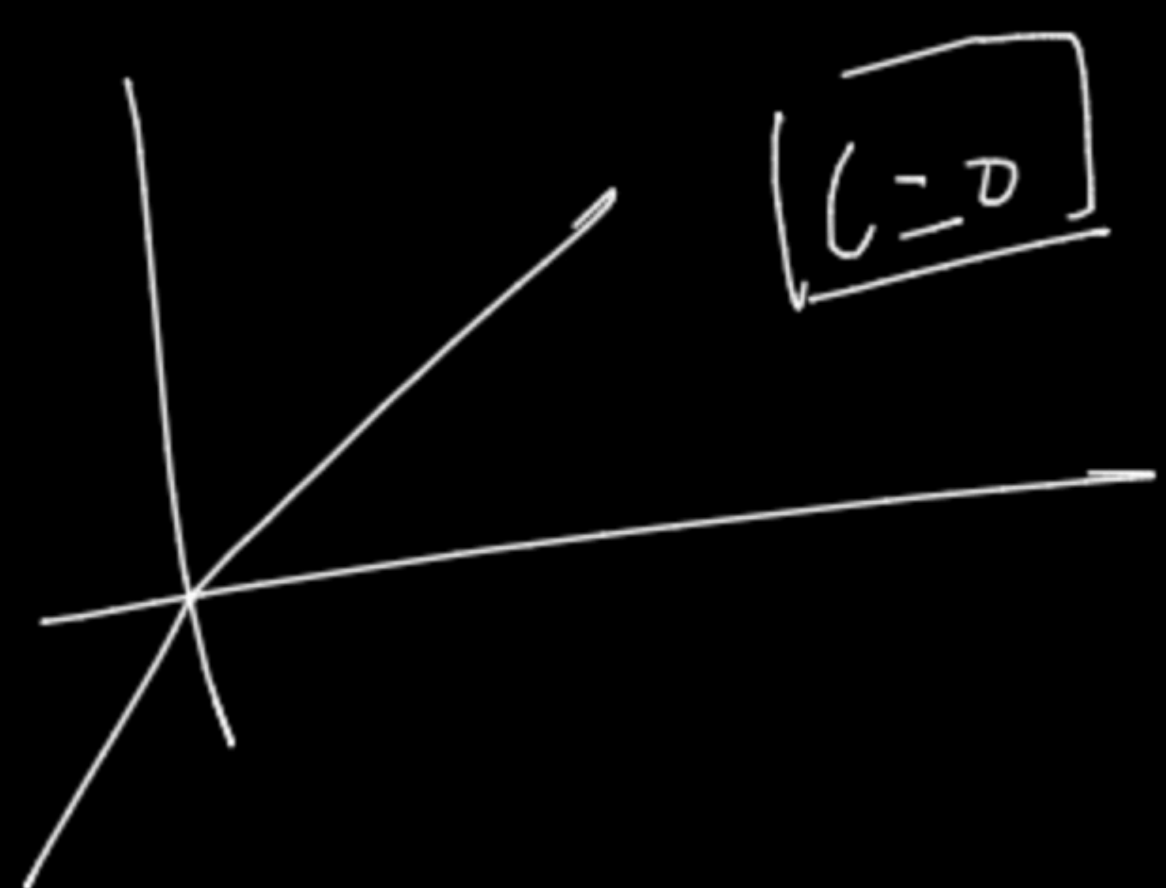
$\Rightarrow$ den^n

(Case i)



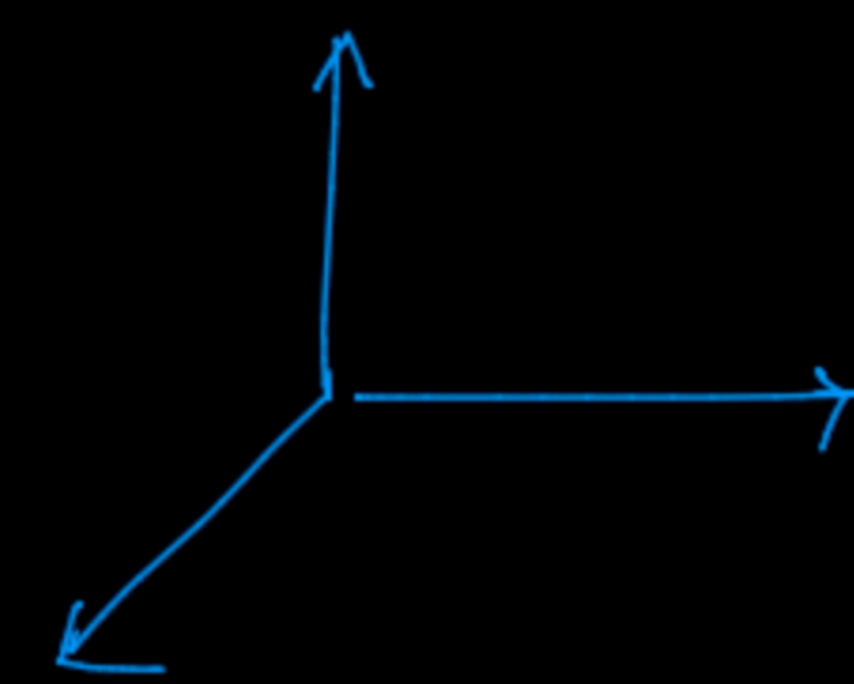
$\rightarrow$ eq^n $\sum_{i=1}^{n} w^T x_i + w_0 = 0$

(Case ii)

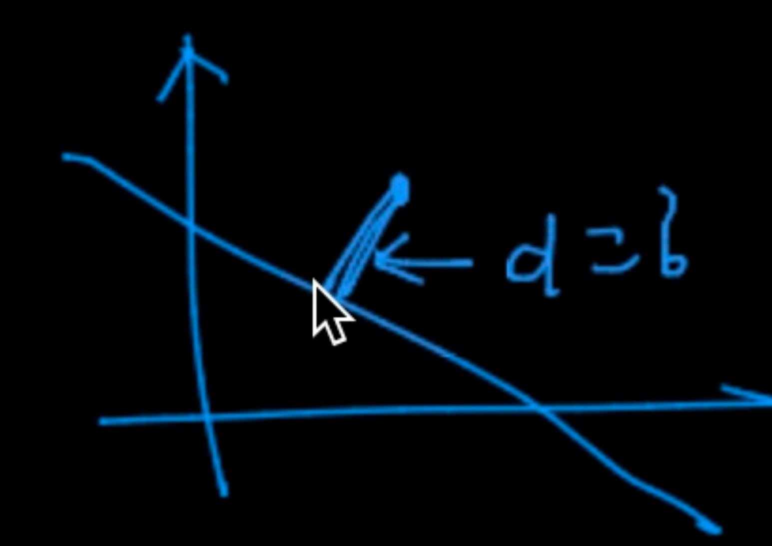


$$\boxed{\sum_{i=1}^{n} w^T x_i = 0}$$

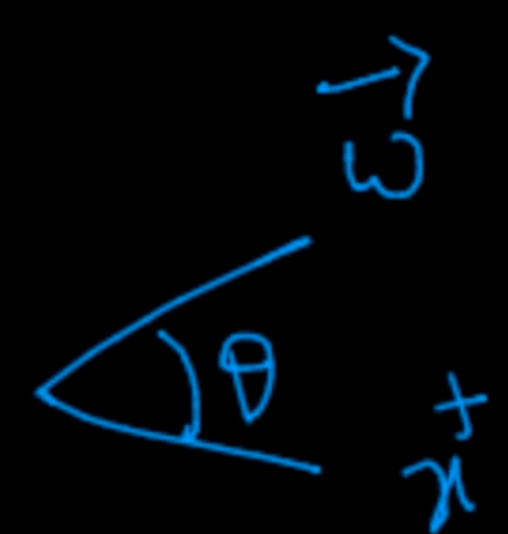
$\Pi_n \quad w^T x = 0$



$$w = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} \qquad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}$$



$$w \cdot x = w^T \cdot x = \|w\| \|x\| \cdot \cos\theta$$

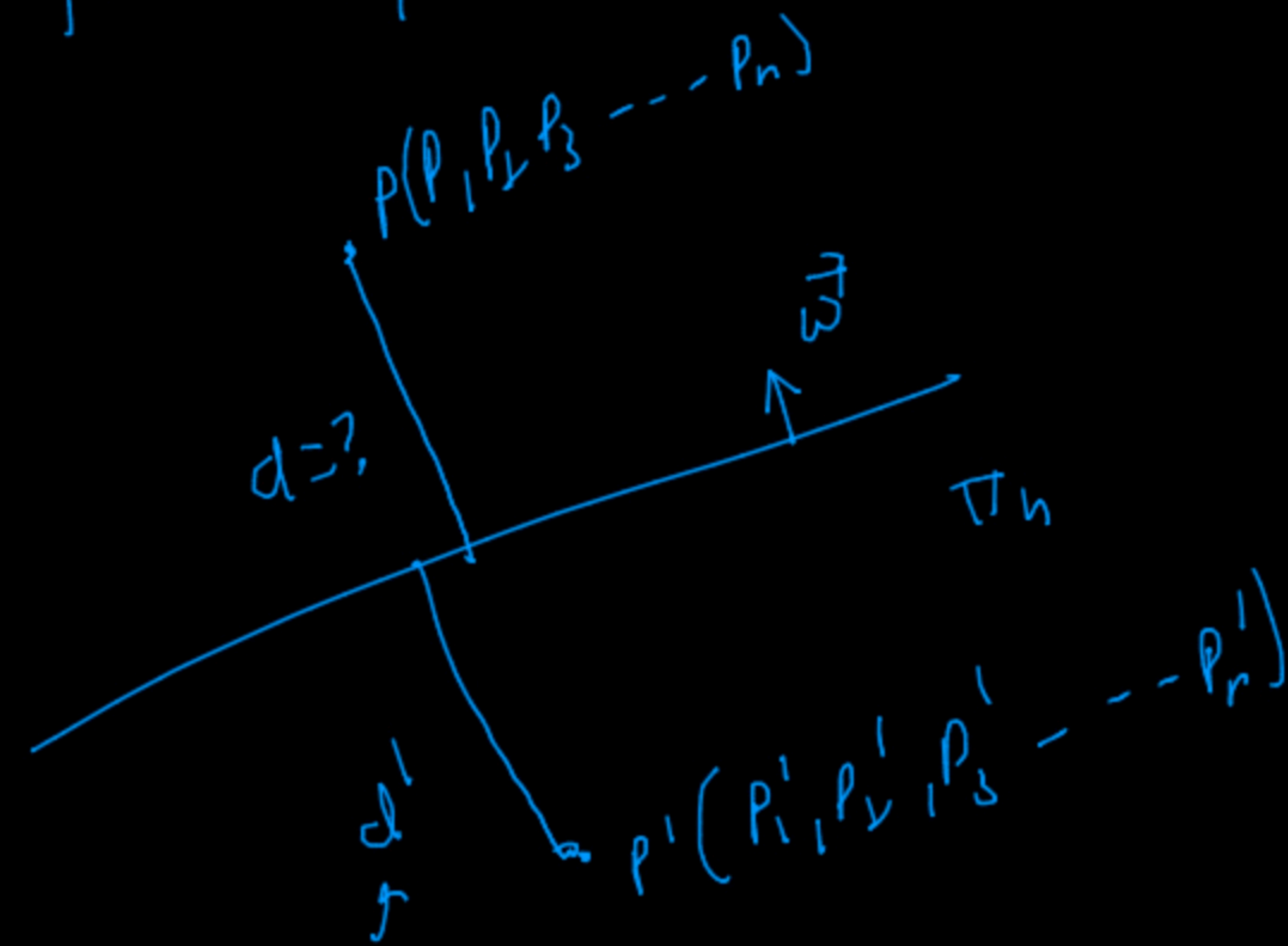


90° ↑

$$\boxed{w \cdot x = 0} \quad \text{if} \quad w \perp \Pi$$

Distance of a points from a plane

Log Reg
Lin



+ve

+w

$w^T x = 0$

$$d = \frac{w^T * p}{||w||}$$

$$d = \frac{w \cdot p}{||w||} \rightarrow +ve$$

Here $\vec{w}$ & p are in same side

$$d' = \frac{w \cdot p'}{||w||} \rightarrow -ve$$

ML + DL

Advanced

Basic